

Reservea — Reservation & Booking Orchestrator (MVP)

A spec-driven MVP that orchestrates the full booking lifecycle — **search → hold → pay → confirm → cancel/refund** — with a MakeMyTrip-inspired React SPA and a hexagonal FastAPI backend.

Feature: 001-reservation-orchestrator — see [specs/001-reservation-orchestrator/](#) for the spec, plan, and tasks.

Tech stack

Layer	Stack
Backend	Python 3.12, FastAPI, Pydantic v2, SQLAlchemy 2.x async, aiosqlite
Frontend	React 18, Vite, TypeScript (strict), Tailwind CSS
Persistence	SQLite (data/reservations.db), seeded from reservation_data.json
MCP	src/mcp_payment_server.py — authorize / capture / refund tools
Tests	pytest (unit / integration / contract / refund_gate)

Prerequisites

- Python 3.12
 - Node.js 18+ and npm
 - (optional) git
-

Quick setup

1. Backend

```
python -m venv .venv
```

```
.venv\Scripts\Activate.ps1
```

```
pip install -e ".[dev]"
```

2. Frontend

cd frontend

npm install

cd ..

Running the app

Backend (auto-seeds data\reservations.db on first run)

.venv\Scripts\python.exe -m src.main *# http://localhost:8000 — OpenAPI at /docs*

Frontend (in a second terminal)

cd frontend

npm run dev *# http://localhost:5173*

Payment MCP server (optional)

python -m src.mcp_payment_server *# stdio*

python -m src.mcp_payment_server --sse *# SSE on :8001*

Stopping servers

Stop-Process -Id (Get-NetTCPConnection -LocalPort 8000).OwningProcess -Force

Stop-Process -Id (Get-NetTCPConnection -LocalPort 5173).OwningProcess -Force

Testing

pytest -q

pytest -m refund_gate *# Constitution Principle VI — 100% coverage target*

Useful endpoints

- GET /api/guests
- POST /api/search
- POST /api/reservations — place hold

- POST /api/reservations/{id}/pay
 - POST /api/reservations/{id}/confirm
 - POST /api/reservations/{id}/refunds
 - GET /api/reservations — list all
 - OpenAPI docs: <http://localhost:8000/docs>
-

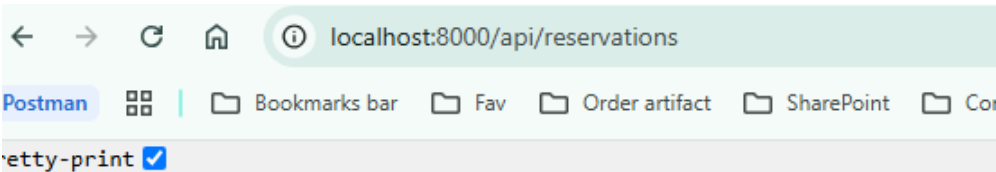
UI Backend

C:\WINDOWS\system32\cmd. X

TCP	[::1]:5173	[::1]:60965	ESTABLISHED	18380
TCP	[::1]:5173	[::1]:63275	ESTABLISHED	18380
TCP	[::1]:60965	[::1]:5173	ESTABLISHED	12692
TCP	[::1]:63275	[::1]:5173	ESTABLISHED	15640

(.venv) C:\Users\akuma405\Documents\P1_Reservation_Booking_ORCH>.venv\Scripts\python.exe -m src.main

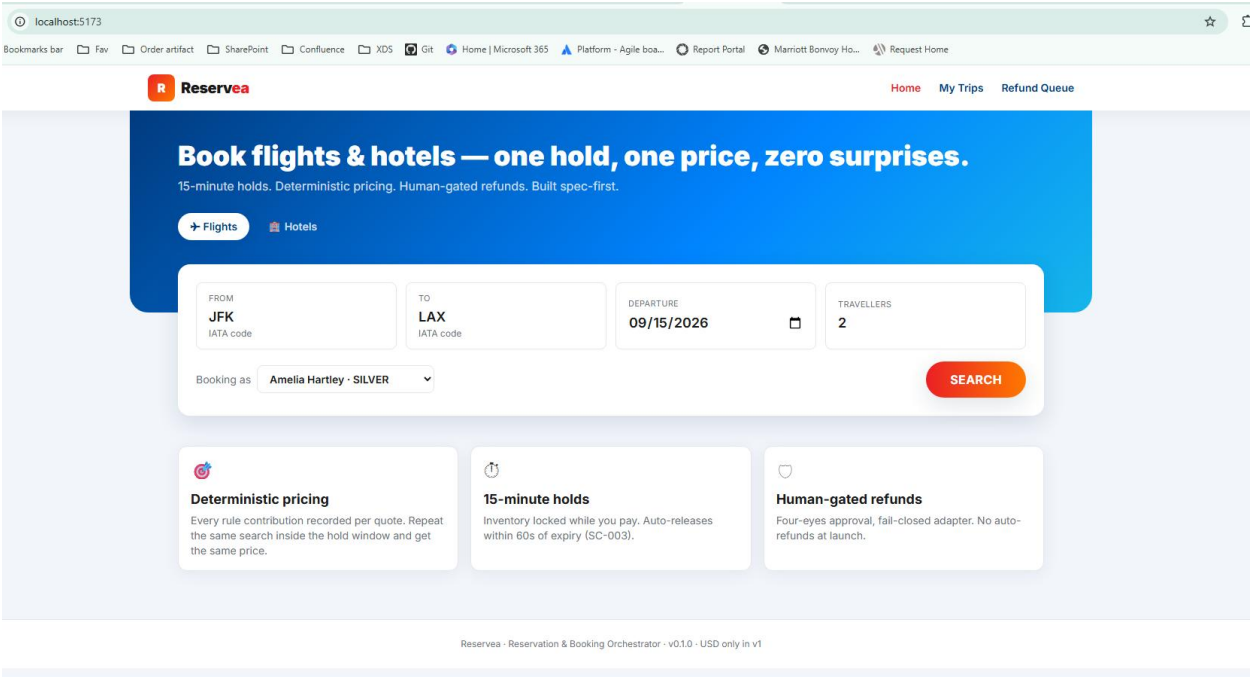
```
INFO: Started server process [27180]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: 127.0.0.1:60287 - "GET /api/reservations HTTP/1.1" 200 OK
INFO: 127.0.0.1:55839 - "GET /api/guests HTTP/1.1" 200 OK
INFO: 127.0.0.1:59218 - "POST /api/search HTTP/1.1" 200 OK
INFO: 127.0.0.1:49194 - "GET /api/guests HTTP/1.1" 200 OK
INFO: 127.0.0.1:54606 - "POST /api/search HTTP/1.1" 200 OK
INFO: 127.0.0.1:56789 - "POST /api/reservations HTTP/1.1" 201 Created
INFO: 127.0.0.1:58700 - "GET /api/reservations/R-AD7E659C4E HTTP/1.1" 200 OK
INFO: 127.0.0.1:64474 - "GET /api/reservations/R-AD7E659C4E HTTP/1.1" 200 OK
INFO: 127.0.0.1:49743 - "GET /api/reservations/R-AD7E659C4E HTTP/1.1" 200 OK
INFO: 127.0.0.1:59981 - "POST /api/reservations/R-AD7E659C4E/pay HTTP/1.1" 200 OK
INFO: 127.0.0.1:54702 - "GET /api/reservations/R-AD7E659C4E HTTP/1.1" 200 OK
INFO: 127.0.0.1:58507 - "GET /api/reservations/R-AD7E659C4E HTTP/1.1" 200 OK
INFO: 127.0.0.1:57349 - "POST /api/reservations/R-AD7E659C4E/confirm HTTP/1.1" 200 OK
INFO: 127.0.0.1:61422 - "GET /api/reservations HTTP/1.1" 200 OK
INFO: 127.0.0.1:60737 - "GET /api/guests HTTP/1.1" 200 OK
INFO: 127.0.0.1:63726 - "GET /api/reservations HTTP/1.1" 200 OK
INFO: 127.0.0.1:60151 - "GET /api/guests HTTP/1.1" 200 OK
INFO: 127.0.0.1:59196 - "POST /api/search HTTP/1.1" 200 OK
INFO: 127.0.0.1:63103 - "POST /api/reservations HTTP/1.1" 201 Created
INFO: 127.0.0.1:54812 - "GET /api/reservations/R-40F37A9347 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60753 - "GET /api/reservations/R-40F37A9347 HTTP/1.1" 200 OK
INFO: 127.0.0.1:57587 - "POST /api/reservations/R-40F37A9347/pay HTTP/1.1" 200 OK
INFO: 127.0.0.1:55893 - "GET /api/reservations/R-40F37A9347 HTTP/1.1" 200 OK
INFO: 127.0.0.1:59353 - "POST /api/reservations/R-40F37A9347/confirm HTTP/1.1" 200 OK
INFO: 127.0.0.1:57345 - "GET /api/reservations HTTP/1.1" 200 OK
INFO: 127.0.0.1:63814 - "GET /api/reservations/R-40F37A9347 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60170 - "GET /api/reservations/R-40F37A9347 HTTP/1.1" 200 OK
INFO: 127.0.0.1:52194 - "GET /api/reservations HTTP/1.1" 200 OK
INFO: 127.0.0.1:60844 - "GET /api/reservations/R-40F37A9347 HTTP/1.1" 200 OK
INFO: 127.0.0.1:63843 - "GET /api/reservations/R-40F37A9347 HTTP/1.1" 200 OK
INFO: 127.0.0.1:56028 - "GET /api/reservations/R-40F37A9347 HTTP/1.1" 200 OK
INFO: 127.0.0.1:57460 - "GET /api/refunds HTTP/1.1" 200 OK
INFO: 127.0.0.1:58215 - "GET /api/reservations HTTP/1.1" 200 OK
INFO: 127.0.0.1:53276 - "GET /api/guests HTTP/1.1" 200 OK
INFO: 127.0.0.1:54953 - "POST /api/search HTTP/1.1" 200 OK
INFO: 127.0.0.1:58820 - "POST /api/reservations HTTP/1.1" 201 Created
INFO: 127.0.0.1:59693 - "GET /api/reservations/R-37AAC84C93 HTTP/1.1" 200 OK
INFO: 127.0.0.1:53589 - "POST /api/reservations/R-37AAC84C93/pay HTTP/1.1" 200 OK
INFO: 127.0.0.1:59571 - "GET /api/reservations/R-37AAC84C93 HTTP/1.1" 200 OK
INFO: 127.0.0.1:65219 - "POST /api/reservations/R-37AAC84C93/confirm HTTP/1.1" 200 OK
INFO: 127.0.0.1:61260 - "GET /api/reservations HTTP/1.1" 200 OK
```



```
{
  "reservation_id": "BK-003",
  "guest_id": "G-002",
  "item_type": "FLIGHT",
  "item_id": "FL-005",
  "state": "HELD",
  "total_amount": 129.99,
  "currency": "USD",
  "confirmation_code": null,
  "check_in": null,
  "check_out": null,
  "nights": null,
  "hold_expires_at": "2026-09-14T08:15:00",
  "price_breakdown": [],
  "created_at": "2026-09-14T08:00:00",
  "updated_at": "2026-09-01T09:20:59.366893",
  "version": 1
},
{
  "reservation_id": "BK-004",
  "guest_id": "G-003",
  "item_type": "ROOM",
  "item_id": "RM-003",
  "state": "CANCELLED",
  "total_amount": 393,
  "currency": "USD",
  "confirmation_code": "PMT-004",
  "check_in": "2026-09-18T00:00:00",
  "check_out": "2026-09-20T00:00:00",
  "nights": 2,
  "hold_expires_at": null,
  "price_breakdown": [],
  "created_at": "2026-09-10T22:40:00",
  "updated_at": "2026-09-01T09:20:59.366893",
  "version": 1
},
{
  "reservation_id": "BK-002",
  "guest_id": "G-004",
  "item_type": "ROOM",
  "item_id": "RM-010",
  "state": "CONFIRMED",
  "total_amount": 1267,
  "currency": "USD",
  "confirmation_code": "PMT-002",
  "check_in": "2026-09-15T00:00:00",
  "check_out": "2026-09-20T00:00:00",
  "nights": 5,
  "hold_expires_at": null,
  "price_breakdown": [],
  "created_at": "2026-09-02T12:30:00",
  "updated_at": "2026-09-01T09:20:59.366893",
  "version": 1
},
{
  "reservation_id": "R-40F37A9347",
  "guest_id": "G-001",
  "item_type": "ROOM",
  "item_id": "RM-001",
  "state": "CONFIRMED",
  "total_amount": 1267,
  "currency": "USD",
  "confirmation_code": "PMT-001",
  "check_in": "2026-09-15T00:00:00",
  "check_out": "2026-09-20T00:00:00",
  "nights": 5,
  "hold_expires_at": null,
  "price_breakdown": [],
  "created_at": "2026-09-02T12:30:00",
  "updated_at": "2026-09-01T09:20:59.366893",
  "version": 1
}
```

UI screenshots

Home — search form



Search results — offers

localhost:5173/search?kind=ROOM&guests=2&guest_id=G-001&location=Los+Angeles&check_in=2026-09-15&check_out=2026-09-18

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R

Reservea

HomeMy TripsRefund Queue

← New search · Los Angeles · 2026-09-15 → 2026-09-18

ROOM

JW Marriott — Standard King

Los Angeles, CA · 3 nights · fits 2

Price breakdown (2 rules)

HOLD FOR 15 MIN

Total
\$571.82
USD

ROOM

JW Marriott — Deluxe King

Los Angeles, CA · 3 nights · fits 2

Price breakdown (2 rules)

HOLD FOR 15 MIN

Total
\$775.52
USD

ROOM

JW Marriott — Executive Suite

Los Angeles, CA · 3 nights · fits 3

Price breakdown (2 rules)

HOLD FOR 15 MIN

Total
\$1241.12
USD

Reservea · Reservation & Booking Orchestrator · v0.1.0 · USD only in v1

Checkout — hold + countdown + pay

localhost:5173/checkout/R-688C6262FD

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R

Reservea

HomeMy TripsRefund Queue

Checkout

Reservation R-688C6262FD

Item

ROOM · RM-003

Check-in

Tue Sep 15 2026

PRICE BREAKDOWN

base — 3 × 196.50 USD

loyalty_tier — SILVER -3%

Total

\$571.82 USD

State

HELD

Check-out

Fri Sep 18 2026

HOLD EXPIRES IN

00:00

Pay before the timer runs out.

PAY NOW

Cancel hold

Reservea · Reservation & Booking Orchestrator · v0.1.0 · USD only in v1

Confirmation

localhost:5173/confirmation/R-688C6262FD

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R

Reservea

HomeMy TripsRefund Queue

✓

You're booked!

Your reservation is confirmed. A confirmation email has been queued.

RESERVATION
R-688C6262FD

STATE
PAID

ITEM
ROOM · RM-003

TOTAL PAID
\$571.82 USD

My Trips

BOOK ANOTHER

Reservea - Reservation & Booking Orchestrator - v0.1.0 · USD only in v1

Trips

localhost:5173/trips

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R

Reservea

HomeMy TripsRefund Queue

My Trips

All reservations across all guests (demo mode).

BK-003
FLIGHT · FL-005
Guest G-002 · v1 · 9/14/2026, 8:00:00 AM

TOTAL

\$129.99 USD

HELD

Details

Cancel

BK-004
ROOM · RM-003
Guest G-003 · v1 · 9/10/2026, 10:40:00 PM
Confirmation: PMT-004

TOTAL

\$393.00 USD

CANCELLED

Details

BK-002
ROOM · RM-010
Guest G-004 · v2 · 9/2/2026, 12:30:00 PM
Confirmation: PMT-002

TOTAL

\$1267.00 USD

CANCELLED

Details

R-688C6262FD
ROOM · RM-003
Guest G-001 · v3 · 9/2/2026, 4:42:44 AM
Confirmation: 22B8RMB3

TOTAL

\$571.82 USD

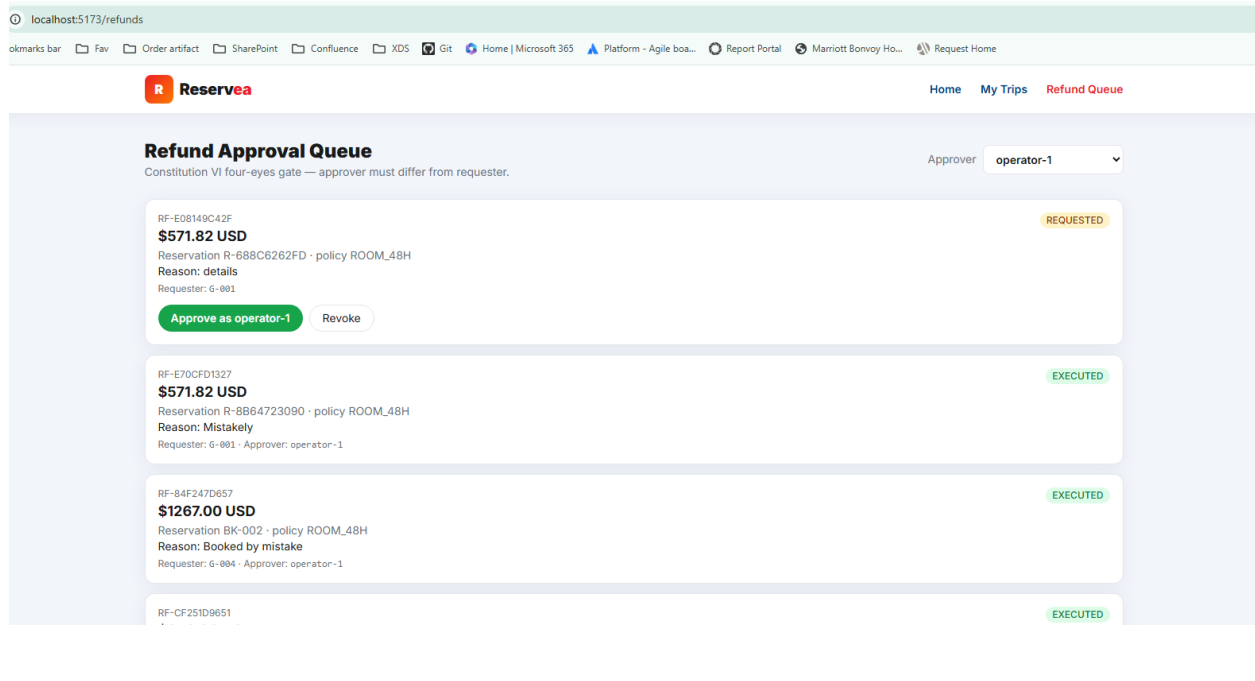
CONFIRMED

Details

Refund reason

Request refund

Refund queue (four-eyes approval)



Folder structure

P1_Reservation_Booking_ORCH/

- └─ .github/
- | └─ agents/ # Custom Copilot agents (Reservation_Search_Agent, ...)
- | └─ skills/ # Speckit skill definitions
- └─ .specify/ # Spec-Kit templates & scripts
- └─ .vscode/
- | └─ mcp.json # MCP server registrations (payment-mcp)
- └─ data/
- | └─ reservations.db # SQLite DB (auto-created on first run)
- └─ docs/
- | └─ screenshots/ # UI screenshots referenced by this README
- └─ frontend/ # React 18 + Vite + TS + Tailwind SPA
- | └─ index.html

- | └─ package.json
- | └─ tailwind.config.js
- | └─ postcss.config.js
- | └─ tsconfig.json
- | └─ vite.config.ts # dev server on :5173, proxies /api → :8000
- | └─ src/
 - | └─ App.tsx
 - | └─ main.tsx
 - | └─ api.ts # typed API client
 - | └─ index.css # Tailwind entry
 - | └─ routes/
 - | └─ Home.tsx # search form
 - | └─ SearchResults.tsx # offer cards
 - | └─ Checkout.tsx # hold + countdown + pay
 - | └─ Confirmation.tsx # booking confirmed
 - | └─ Trips.tsx # guest trips
 - | └─ RefundQueue.tsx # 4-eyes refund approvals
- | └─ specs/
 - | └─ 001-reservation-orchestrator/ # spec.md, plan.md, tasks.md, contracts/
- | └─ src/ # Backend (hexagonal)
 - | └─ main.py # ASGI entrypoint (FastAPI app)
 - | └─ config.py
 - | └─ mcp_payment_server.py # MCP: authorize / capture / refund
 - | └─ api/
 - | └─ routes.py
 - | └─ schemas.py

- | | └─ deps.py
- | └─ application/ # Use cases / saga
- | | └─ orchestrator.py # search → hold → pay → confirm → refund
- | | └─ search_service.py
- | └─ domain/ # Pure Python — no framework imports
- | | └─ policies/
- | | └─ pricing/
- | | └─ state_machines/
- | | └─ value_objects/
- | └─ adapters/ # Hexagonal adapters
- | | └─ persistence/
- | | | └─ database.py # async engine / session
- | | | └─ models.py # SQLAlchemy ORM
- | | | └─ seeder.py # loads reservation_data.json
- | | └─ providers/
- | | └─ stubs.py # stub inventory / payment providers
- | └─ workers/
- | └─ hold_sweeper.py # expires stale holds
- └─ tests/
- | └─ conftest.py
- | └─ unit/
- | └─ integration/
- └─ reservation_data.json # seed data (guests, rooms, flights, bookings)
- └─ pyproject.toml
- └─ README.md

Acceptance Criteria

AC-1 → FR-002, User Story 1 AC 1

AC-2 → FR-004 (season → occupancy surge → LOS discount → tier discount)

AC-3 → FR-007, FR-008, User Story 1 AC 2

AC-4 → FR-009, Edge Case #2, SC-007

AC-5 → FR-010, FR-011, User Story 4 AC 1, SC-003

AC-6 → FR-016, FR-018, User Story 2 AC 2

Test Case Pass

```
• PS C:\Users\akuma405\Documents\P1_Reservation_Booking_ORCH> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ;
& c:\Users\akuma405\Documents\P1_Reservation_Booking_ORCH\.venv\Scripts\Activate.ps1)
• (.venv) PS C:\Users\akuma405\Documents\P1_Reservation_Booking_ORCH> pytest
===== test session starts =====
platform win32 -- Python 3.12.1, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\akuma405\Documents\P1_Reservation_Booking_ORCH
configfile: pyproject.toml
testpaths: tests
plugins: anyio-4.14.2, asyncio-1.4.0, cov-7.1.0, respx-0.23.1
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 29 items

tests\integration\test_lifecycle.py ..... [ 17%]
tests\unit\domain\test_cancellation_policy.py ..... [ 34%]
tests\unit\domain\test_pricing.py ..... [ 55%]
tests\unit\domain\test_refund_gate.py ..... [100%]

===== 29 passed in 9.99s =====
• (.venv) PS C:\Users\akuma405\Documents\P1_Reservation_Booking_ORCH> []
```

Notes

- The DB is created and seeded at data/reservations.db from reservation_data.json on backend startup.
- Refund approvals enforce a four-eyes rule: approver must be != requester.
- Idempotency keys and an outbox/audit pattern are implemented for durability.
- Hold expiry is enforced by src/workers/hold_sweeper.py; the Checkout page shows a live countdown driven by reservation.hold_expires_at.

MCP Server Tools

#	Tool	Purpose	Parameters	Returns
1	mcp_payment_server.py:82	Place a 15-min hold on an offer and authorize	offer_id, guest_id, payment_token, check_in?, check_out?	reservation_id, state (PAID or CANCELLED_PAYMENT_FAILED), total_amount, current_amount

#	Tool	Purpose	Parameters	Returns
		payment in one step (equivalent to POST /api/reservations then POST /api/reservations/{id}/pay).		cy, hold_expires_at, provider_ref, correlation_id
2	mcp_payment_server.py:170	Capture funds and confirm a previously authorized (PAID) reservation (equivalent to POST /api/reservations/{id}/confirm).	reservation_id, guest_id	reservation_id, state (CONFIRMED), confirmation_code, total_amount, currency, correlation_id
3	mcp_payment_server.py:251	Cancel reservation + create refund (FR-018 policy) + approve it via four-eyes gate (approver_sub != guest_id).	reservation_id, guest_id, reason, approver_sub (default "operator-1")	refund_id, reservation_id, state (EXECUTED), amount, currency, policy_code, reason, requester_sub, approver_sub, correlation_id

Custom Agent

The screenshot shows a VS Code editor with a custom agent configuration for 'Reservation_Search_Agent'. The configuration includes a description, argument hint, tools (search), and usage instructions. The chat window on the right shows a query for bookings for guest G-002 and provides a table of results.

Reservation_Search_Agent Configuration:

- name:** Reservation_Search_Agent
- description:** Describe what this custom agent does and when to use it.
- argument-hint:** The inputs this agent expects, e.g., "a task to implement" or "a question to answer".
- tools:** search

When to use:

- Use this agent instead of the default when the task is: querying, diagnosing, or improving the search/offer flows in the repository; interacting with local dev servers (backend API on :8000, frontend on :5173); writing or updating UI/API code for search results; writing tests that exercise search behavior.

Persona & Role:

- Role: pragmatic developer-assistant tuned for reservation-search tasks.
- Tone: concise, direct, and collaborative — provide minimal changes and concrete commands.

Allowed tools & preferences:

- Prefer: `read_file`, `file_search`, `grep_search`, `apply_patch`, `run_in_terminal`, `open_browser_page`, `view_image`, `read_notebook_cell_output` (when present).
- Use `run_in_terminal` for running local servers, tests, or quick API calls only after listing the exact commands to run.
- Use `apply_patch` for source changes: prefer minimal, well-scoped edits with tests when possible.
- Avoid: external web requests, publishing artifacts, or changes that require secrets (database passwords, private keys). If a secret is required, ask the

Chat Window:

ENGINEERING CONSTITUTION FOR BOOKING PLATFORM

If the backend isn't running, start it first:

```
./venv\Scripts\python.exe -m src.main
```

Paste the output and I'll summarize the bookings for G-002.

Search for G-002 and provide the output

Searched for G-002 and reviewed reservation_data.json

Bookings for guest G-002 (Rohan Mehta) from seeded data (from reservation_data.json):

ID	Type	Item	Status	Amount	Dates	Payment
BK-003	T	005	PENDING	\$129.99	expires 2026-09-14T08:15:00Z	---

There is one prior CONFIRMED room booking earlier in the file for another guest (BK-002 is for G-001 — Amelia). G-002 has 1 seeded booking (BK-003, PENDING, expired hold).

Note: this is from the seed file. If additional bookings were created via the running API, you'd need to hit `GET /api/reservations` (terminal is currently disabled) to see them. Want me to draft a small script or endpoint to filter reservations by `guest_id`?

Github repository

https://github.com/ankiteraccenture/P1_Reservation_Booking_ORCH